Beyond a Single Structure: Exploring Multistate Protein Conformational Landscapes with AI Structure Prediction

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Abstract

Proteins function through conformational ensembles. To assess the ability of current foundational structure inference models to predict multiple protein states, we generated 22,000 monomer models (11,000 per target; 1,000 per sampling method per target) for E. coli Adk (PDB IDs 4AKE/1AKE)[1,2] and Mad2 mitotic spindle assembly checkpoint protein (PDB IDs 1DUJ/1S2H)[3,4] using 11 different Artificial Intelligence Sampling (AIS) methods. Four AIS methods involve applying a stochastic 40% MSA column masking to foundational structure inference models, adapted from Kalakoti and Wallner (Fig. 1a).[5] Shared Cα pair distances from the AIS models were used as features for clustering with UMAP, PCA, t-SNE and Bonsai. These AIS procedures produced models with best TM-scores (US-align) of 0.97/0.99 against open/closed adenylate kinase references (4AKE/1AKE) and 0.80/0.83 against open/closed Mad2 references (1DUJ/1S2H). Models were used as templates with Boltz2 in no-MSA mode to retrieve reported pTM scores, similar to the AFRank methodology proposed by Roney and Ovchinnikov (Fig. 1b).[6] Each target was analyzed on four A100 GPUs in approximately 24 hours. This AIS protocol is continued on a set of 500+ protein targets with experimental evidence of multiple states. All AIS models and Boltz2 template-only refolds for the two targets presented here are deposited on Zenodo:
Adenylate kinase: 10.5281/zenodo.23601146
Mad2: 10.5281/zenodo.23601144
Code Accessibility: https://github.com/benavt/SCHERAGA_POSTER
A virtual poster accompanies this physical poster, accessible by QR code.

Background

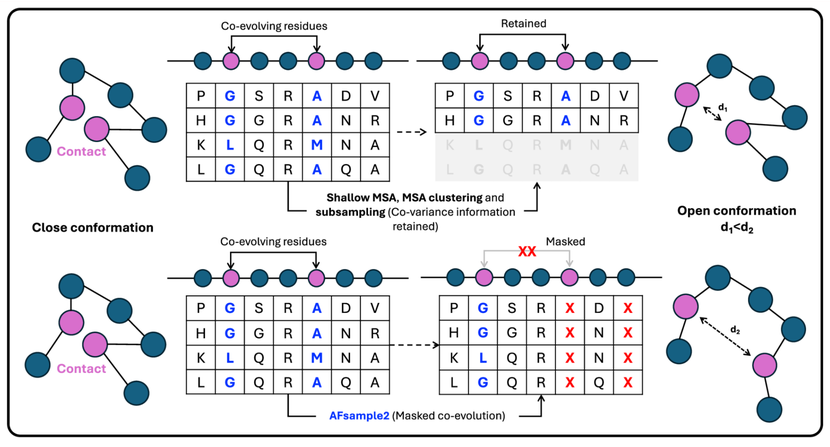


Figure 1a. MSA subsampling and masking. Kalakoti & Wallner (2025), Fig. 1a[5], CC BY 4.0.

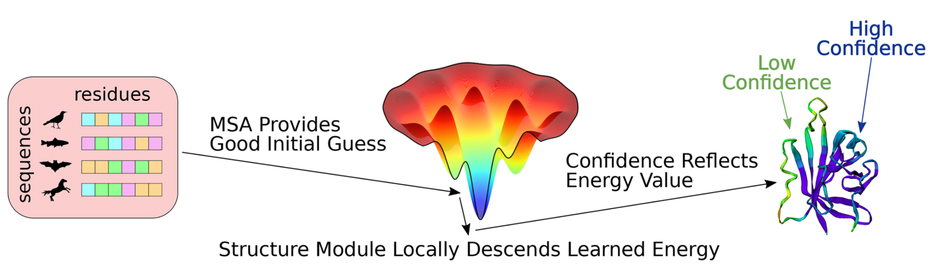


Figure 1b. Learned structure landscape. From Roney & Ovchinnikov (2022), Fig. 1[6]. CC BY 4.0.

A / E. coli adenylate kinase

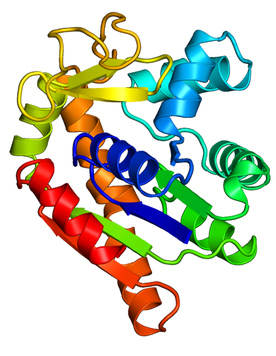
S1 · AlphaFlow-MD
1AKE_alphaflow-md_0703.pdb



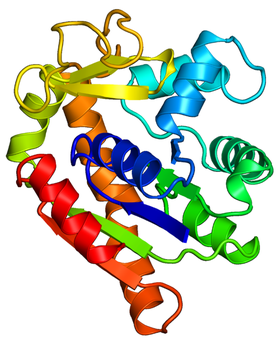
4AKE open



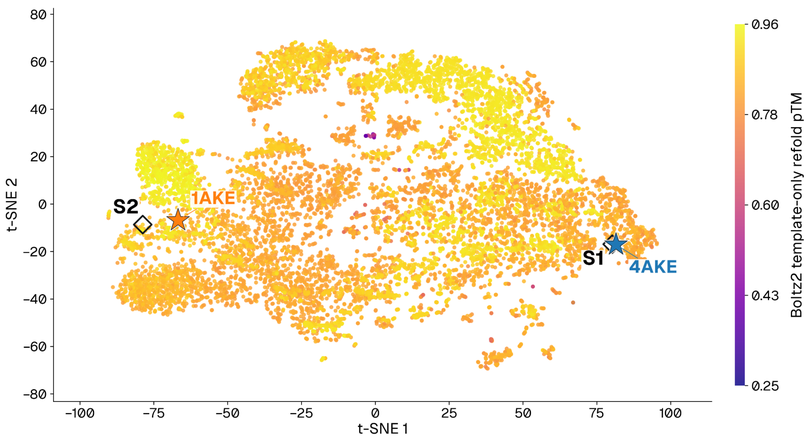
S2 · AlphaFlow-PDB
1AKE_alphaflow-pdb_0327.pdb



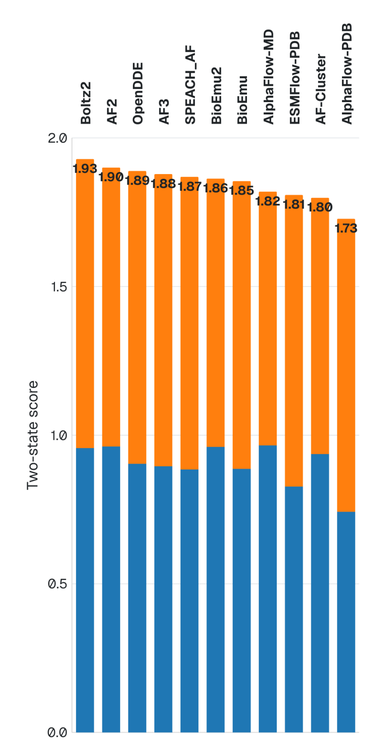
1AKE closed



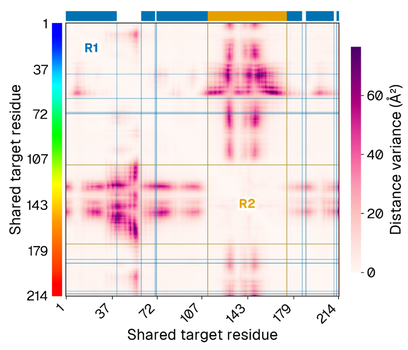
A1. AI sampling produces open and closed states. TM/Boltz2 refold pTM: S1 (4AKE[1]) 0.97/0.82; S2 (1AKE[2]) 0.99/0.88.



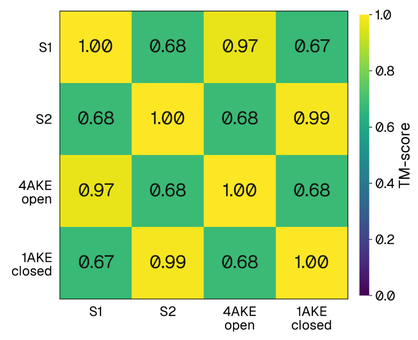
A2. t-SNE projection. 11,000 models; shared pairwise Cα distances, colored by Boltz2 template-only refold pTM. ◇ S1/S2 ★ 4AKE open[1] ★ 1AKE closed[2]



A5. Two-state score. Best TM(■ 4AKE open[1]) + best TM(■ 1AKE closed[2]), over 1,000 models per method. Methods: **BioEmu**, **BioEmu2**, **SPEACH_AF**, **AF-Cluster**, **AlphaFlow-PDB**, **AlphaFlow-MD**, **ESMFlow-PDB**, and 40% stochastic column masking applied to **AF2**, **AF3**, **OpenDDE**, and **Boltz2**.



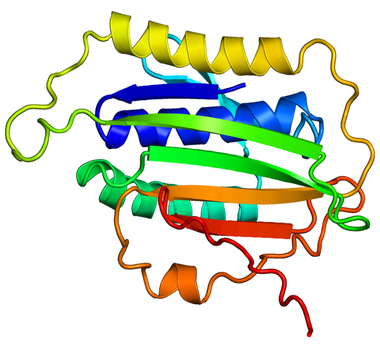
A3. Cα distance variance. R1: core anchor used for structural alignment. R2: A112–173 moving comparison region.



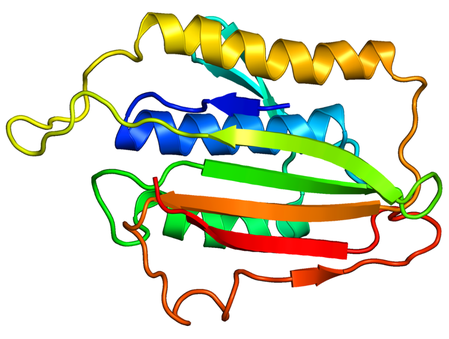
A4. TM-score matrix. S1/S2 and PDB controls. 214 shared Cα. Best AIS TM: 4AKE open[1] 0.97; 1AKE closed[2] 0.99.

B / Mad2 mitotic spindle checkpoint protein

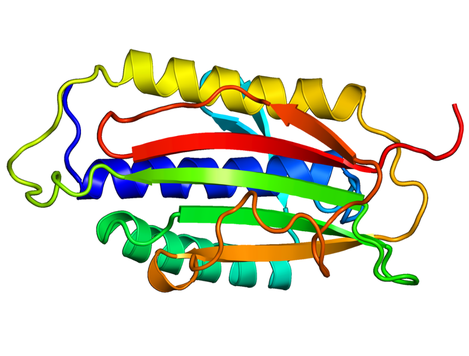
S1 · AF-Cluster
MAD2_afcluster_0675.pdb



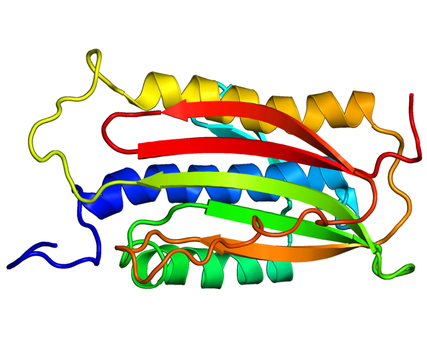
1DUJ open



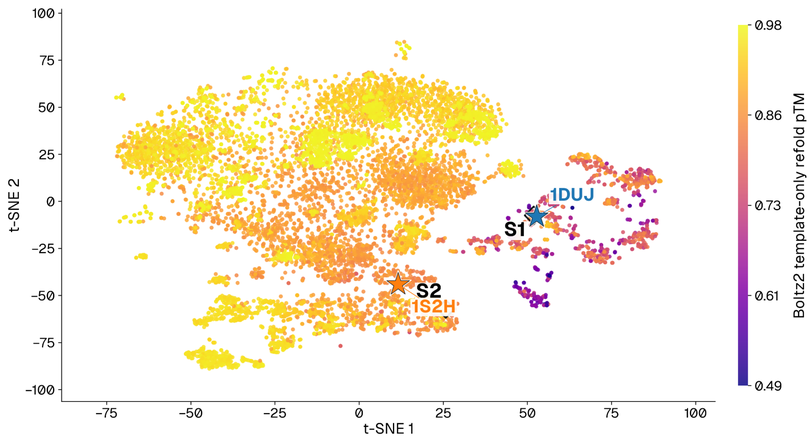
S2 · Boltz2
1KLQ_MAD2_ALONE__Boltz2_Exact_40pct__0852.cif



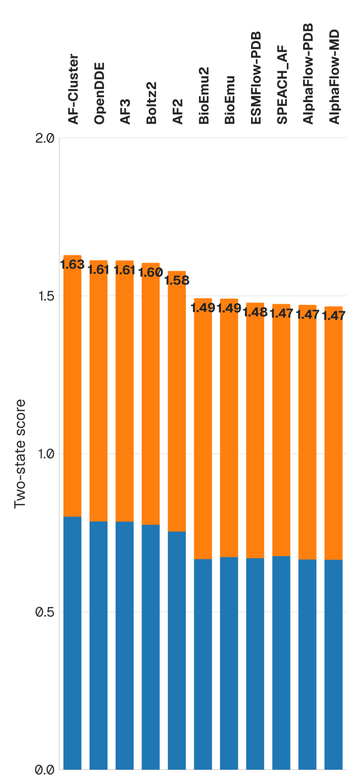
1S2H closed



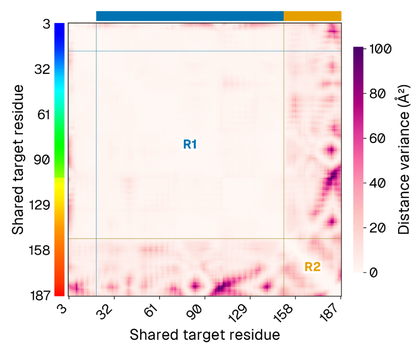
B1. AI sampling produces open and closed states. TM/Boltz2 refold pTM: S1 (1DUJ[3]) 0.80/0.89; S2 (1S2H[4]) 0.83/0.94.



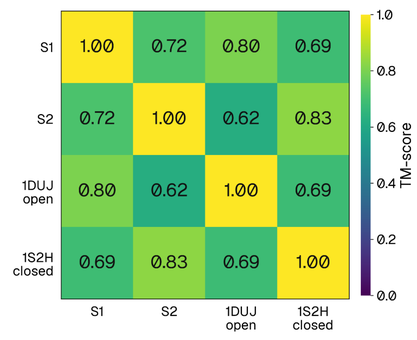
B2. t-SNE projection. 11,000 models; shared pairwise Cα distances, colored by Boltz2 template-only refold pTM. ◇ S1/S2 ★ 1DUJ open[3] ★ 1S2H closed[4]



B5. Two-state score. Best TM(■ 1DUJ open[3]) + best TM(■ 1S2H closed[4]), over 1,000 models per method.



B3. Cα distance variance. R1: core anchor used for structural alignment. R2: shared C-terminal residues A151–187.



B4. TM-score matrix. S1/S2 and PDB controls. 175 shared Cα. Best AIS TM: 1DUJ open[3] 0.80; 1S2H closed[4] 0.83.

References

Additional references can be found in the virtual poster.

1. Müller CW, Schlauderer GJ, Reinstein J, Schulz GE (PDB ID: 4AKE). Adenylate kinase motions during catalysis: an energetic counterweight balancing substrate binding. Structure 4 147–156 (1996). doi:10.1016/S0969-2126(96)00018-4
2. Müller CW, Schulz GE (PDB ID: 1AKE). Structure of the complex between adenylate kinase from Escherichia coli and the inhibitor Ap5A refined at 1.9 Å resolution. A model for a catalytic transition state. Journal of Molecular Biology 224 159–177 (1992). doi:10.1016/0022-2836(92)90582-5
3. Luo X, Fang G, Coldiron M, Lin Y, Yu H, Kirschner MW, Wagner G (PDB ID: 1DUJ). Structure of the Mad2 spindle assembly checkpoint protein and its interaction with Cdc20. Nature Structural Biology 7 224–229 (2000). doi:10.1038/73338
4. Luo X, Tang Z, Xia G, Wassmann K, Matsumoto T, Rizo J, Yu H (PDB ID: 1S2H). The Mad2 spindle checkpoint protein has two distinct natively folded states. Nature Structural & Molecular Biology 11 338–345 (2004). doi:10.1038/nsmb748
5. Kalakoti Y, Wallner B. AFsample2 predicts multiple conformations and ensembles with AlphaFold2. Communications Biology 8 373 (2025). github.com/wallnerlab/AFsample2
6. Roney JP, Ovchinnikov S. State-of-the-art estimation of protein model accuracy using AlphaFold. Physical Review Letters 129 238101 (2022). github.com/jproney/AF2Rank

Acknowledgments

This research sponsored in part by NIH/NIGMS **T32GM141865** and NIH/NIA **R25AG088409.**
Thanks goes also to members of the **Montelione Lab** for useful insight and discussion: **T.A Ramelot**, **Y.J. Huang**, **B.A. Shurina**, and **L.E. Spaman**.
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